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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Software engineer with 3 years of industry experience building and shipping **Python** systems end to end: **FastAPI microservices in Docker**, PostgreSQL data platforms and internal **machine learning applications on AWS**, deployed through a **Jenkins and GitLab CI/CD pipeline**. Strong Python with hands-on **machine learning framework** work across LLM applications, RAG pipelines and a **multi-architecture deep learning study** (CNN, ViT and ResNet-18 encoders with LSTM and attention decoders, benchmarked over five seeds against pre-trained vision-language models). Grounded in **performance work**: cut a calculation bottleneck by adding a second API during development, chose and self-hosted models on **cost and latency** grounds, and evaluated architectures on measured metrics rather than intuition. Delivered **cross-team projects** with Compliance, Client Service, Investment and the security team, took **work packages** from design through release, and mentored up to 4 interns as **de facto team lead**. MSc Data Science (University of Bristol, awaiting result). Seeking to bring that engineering and ML performance experience to Arm's machine learning software stack, and to grow into compiler and C++ work alongside it.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Python Engineering & ML Services

- **Designed and built internal applications end to end** for Compliance, Client Service and Investment: a **ReactJS frontend** and **Python FastAPI microservices in Docker (uvicorn)** with the machine learning and RAG logic behind them - initially the only engineer, later coordinating a frontend developer and up to 4 interns.
- Built **reusable Python components** other applications depend on: SQL-backed tool functions that render fund-comparison PDFs on request, a document **ingestion and embedding pipeline**, and an **LDAP authentication function** powering role-based access control for three user tiers.
- Worked in **short delivery cycles** with Compliance review before each release, **versioned every prompt and configuration change in Git**, and changed the applications on what real usage showed once they were live.

Machine Learning & Retrieval Engineering

- Developed internal **GenAI applications** with **LlamaIndex** and **Azure OpenAI (GPT-3.5 & Llama-3)** answering department-specific questions over internal documents.
- Designed and maintained a **vector database** on **pgvector** with **Azure OpenAI embeddings**, held per department so each team's documents and embeddings stayed siloed and access was enforced.
- Built **agentic tool-calling** where AI agents call **Python functions** as tools to run SQL-backed fund comparisons and generate **PDF** reports on user request.

Performance Evaluation & Model Choices

- **Profiled** the fund-comparison calculation endpoints, found the Python APIs were the bottleneck and **wrote ExpressJS APIs during development** to serve the same calculations faster; the shipping backend stayed Python FastAPI.

- Chose models on **measured cost and latency**: **self-hosted Llama-3 on a GPU EC2 instance** to serve the largest department, kept the stronger **Azure OpenAI** outputs for decision makers, and kept the ability to switch models.
- Switched the email parser from **AWS Textract** to **Azure OpenAI** when a rating agency began sending emails as images - chosen for **robustness and extraction quality** - and scheduled the pipeline (EventBridge 9 AM trigger, Lambda) to **parse over 300 daily emails, saving 60 analyst hours a month**.

Cross-Team Delivery, CI/CD & Leadership

- Delivered **cross-team work packages** with Compliance, Client Service, Investment and the security team, owning milestones from requirements through release and **VAPT sign-off**.
- Built **one-click deployment in Jenkins and GitLab**, a **CI/CD pipeline** shipping the chosen branch to dev or prod, with separate frontend and backend repositories and Docker containers.
- Restricted access to company private IPs with **AWS Security Groups**, added **RBAC against LDAP** and deployed an **Nginx reverse proxy**, one of the fixes that took the applications from **failing VAPT to passing**.
- Acted as **de facto team lead** for 2-4 interns and coordinated the frontend developer across the GenAI team; recognised internally with **Best Team Project in 2024 and 2025**.

PROJECTS

- **University of Bristol MSc Applied Data Science Group Project: VisionCaptioner** (Team project. Multi-phase architecture study comparing **CNN, ResNet-18 and ViT encoders** with symbolic, embedding and **LSTM/attention decoders** across three synthetic datasets of 10,000 images and five seeds, benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca: CellLineSelector** (Team project, co-built. A preprocessing pipeline turning 17 raw inputs into 14 clean joined tables with integrity checks, a per-gene **scoring algorithm** and a RAG layer, built as **five FastAPI services** with a React and TypeScript frontend; ranks around **2,100 cancer cell lines** on target inclusion and exclusion genes, validated with AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **B.Tech Final Year Project: Udacity self-driving car simulator.** (A self-driving car trained using a **Convolutional Neural Network (CNN)** model.)

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby **4th Place** winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon**: Codeplain.ai track **2nd place** winner. (Wingman.ai - An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank**: **3rd Place** winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Git, Docker, Jenkins, GitLab (CI/CD), AWS (EC2 including GPU EC2, S3, ECR, Lambda, EventBridge, SageMaker), Azure OpenAI, Nginx, Linux, Power BI, Claude Code.
- **Languages:** Python (FastAPI, pandas, scikit-learn, LlamaIndex), SQL, NodeJS (Express, ReactJS), R (dplyr, tidyverse, ggplot2), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite